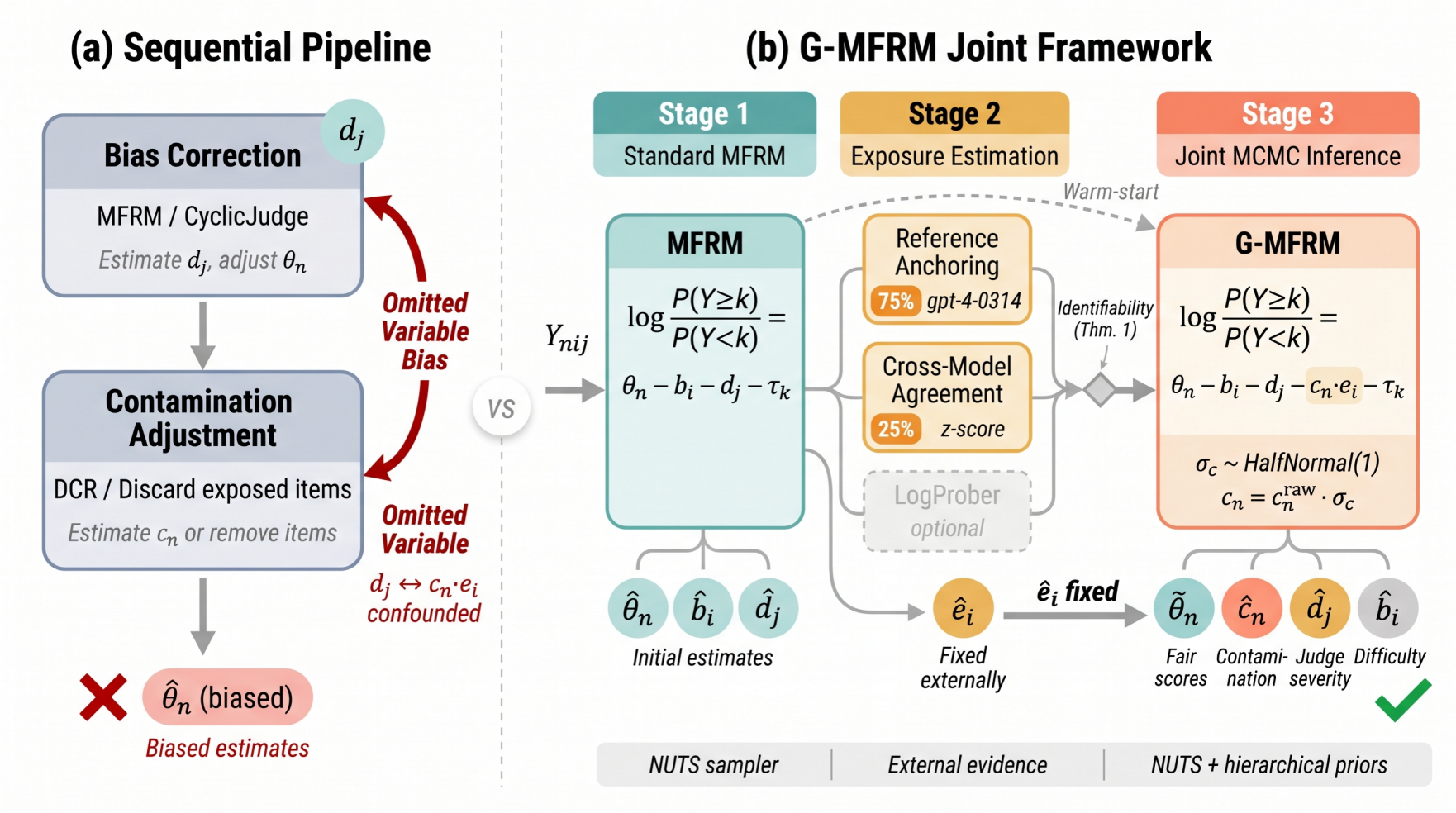

(a) Sequential Pipeline
Bias Correction
$d_j$
MFRM / CyclicJudge
Estimate $d_j$, adjust $\theta_n$
Omitted Variable Bias
Contamination Adjustment
DCR / Discard exposed items
Estimate $c_n$ or remove items
Omitted Variable
$d_j \leftrightarrow c_n \cdot e_i$ confounded
$\hat{\theta}_n$ (biased)
Biased estimates
vs
(b) G-MFRM Joint Framework
Stage 1
Standard MFRM
Stage 2
Exposure Estimation
Stage 3
Joint MCMC Inference
Warm-start
$Y_{nij}$
MFRM
$\log \frac{P(Y \geq k)}{P(Y < k)} = \theta_n - b_i - d_j - \tau_k$
Reference Anchoring
75% gpt-4-0314
Cross-Model Agreement
25% z-score
LogProber
optional
Identifiability (Thm. 1)
G-MFRM
$\log \frac{P(Y \geq k)}{P(Y < k)} = \theta_n - b_i - d_j - c_n \cdot e_i - \tau_k$
$\sigma_c \sim HalfNormal(1)$
$c_n = c_n^{raw} \cdot \sigma_c$
$\hat{\theta}_n$ $\hat{b}_i$ $\hat{d}_j$
Initial estimates
$\hat{e}_i$
Fixed externally
$\hat{e}_i$ fixed
$\tilde{\theta}_n$ $\hat{c}_n$ $\hat{d}_j$ $\hat{b}_i$
Fair scores
Contami-nation
Judge severity
Difficulty
NUTS sampler
External evidence
NUTS + hierarchical priors